

Design Proposal: Spike Investigation - Anomaly Detection & Attribution

Scope: deep on two of the three components - **anomaly detection** and **attribution**. Forecasting is de-scoped: it isn't in the brief's evaluation criteria, and a shallow third pass would dilute the two that matter. All numbers come from output/ after `python run_pipeline.py`.

1. Problem framing

Two distinct ML problems, chained, not merged into one model:

1. **Anomaly detection** - unsupervised time-series outlier detection. No reliable label exists at deployment time, only a business definition of “meaningful,” so the output must be a calibrated departure-from-expectation score, not a classifier.
2. **Attribution** - weakly supervised multi-label classification with unverifiable ground truth. “Why” is never observed, only inferred, and causes co-occur, so a single-label classifier would misrepresent the system's confidence.

Keeping them decoupled matters: different failure modes, different consumers, and merging them hides which half is wrong when the system misfires.

2. Data strategy

`src/data_generator.py`: 8 brands x 6-8 channels (9-channel universe) x 36 months, built as a small structural model rather than noise with labels bolted on:

- **Revenue follows a diminishing-returns response curve** ($\text{spend} \times \text{elasticity}(\text{spend}) \times \text{quality} \times \text{demand} \times \text{noise}$), so a spend cut mechanically raises ROAS and a mix shift toward efficient channels mechanically raises blended ROAS - no special-cased “if artifact then boost ROAS” rule. The artifacts fall out of the economics that produces them in real media data.
- **iROAS comes from a latent quality factor independent of spend level**, so it doesn't move for spend/mix artifacts. That asymmetry is the single strongest attribution feature. It isn't perfectly clean either - deliberately confounded by demand shocks, mirroring real incrementality work.
- **RROI** is an EWMA shrinkage of the attributable-sales ratio toward its trailing 3-month average - “stickier” than ROAS, so a single-month move RROI doesn't confirm is a tell.
- **Leading indicators** (CTR, CPC, PICR, impression share) track the same quality factor as iROAS, not spend/mix - what tells genuine gains apart from budget artifacts.
- **Six cause types** injected as latent-factor shocks, sudden (1mo) or gradual (3mo ramp), often co-occurring with a Dirichlet-split weight. External-demand events are sampled once per category and applied to every brand in that category, so the intended “lifted all boats” signature is genuinely cross-brand. Local events remain brand/channel-specific.
- **Realism knobs**: asymmetric per-brand seasonality, ~2-3% missing periods, one brand launching mid-panel (cold-start case), CPC undefined for TV/OOH (bought on reach). Ground truth is stored **only** for evaluation, never as a feature.

Limitations: one draw from one generative model, encoding our assumptions about response curves, seasonality, and confounding. Survivorship bias is the least faithful mechanism (a reported-metric multiplier, not true placement dropout). Label density (~22%) is higher than a real book of business, to get enough events per cause to evaluate at all.

3. Model design

Anomaly detection. Not mean +/- 2 sigma. Baseline = own trailing median (6mo) + a pooled channel seasonal index, falling back to the same-channel peer trend for brand-channels with fewer than 3 own observations. **sudden_z** (1-month) and **gradual_z** (3-month trailing mean, credited only when the months agree in sign - a lightweight CUSUM) run in parallel off the same residual. Threshold selection minimizes $\text{FP_count} + 5 \times \text{FN_count}$ (raw counts - normalizing by class size pushes the optimum to a degenerate corner); 5:1 reflects that a missed spike leaves the manual process as the only backstop, while a false positive costs minutes. The deployment threshold is **0.50**; evaluation is brand-held-out rather than measured on the calibration rows (Section 5).

The detector is **one-sided** (`direction="up"`). Every cause in the taxonomy raises the headline metric, so scoring on $|z|$ spent half the review budget on *collapses* no cause could explain, and produced incoherent narratives (“spend fell ... consistent with a budget cut” attached to a ROAS drop). Collapses route to the separate performance-decline

review. Analyst narratives report the detector that actually fired: `sudden_z` for sudden alerts and `gradual_z` for gradual alerts.

Rejected: rolling mean/std (not robust to the anomaly pulling its own window); per-series STL (most series too short - as low as 10 months post-launch); a global IsolationForest (loses channel seasonality, nothing an analyst can check against the chart).

Attribution. Independent LightGBM binary classifiers, one per cause (not one softmax) - causes co-occur, so outputs are calibrated marginal likelihoods and deliberately do **not** sum to one. Each group-held-out fold owns its median imputer and small regularized model (60 trees, 7 leaves); an inner event-grouped loop fits Platt calibration without seeing the evaluated event. The serving model averages the calibrated fold ensemble. **Rejected:** normalizing marginals into an “explained-signal share” (not probabilistically meaningful); multi-class softmax; a neural model.

4. Feature engineering

All features are z-scores/ratios against the same baseline machinery used for detection, re-used for `iroas`, `rroi`, `spend`, `ctr`, `cpc`, `picr`, `impression_share`, and channel spend share:

Feature	Separates
<code>roas_iroas_divergence</code> , <code>roas_rroi_divergence</code>	genuine gain (both move) vs. artifact (ROAS only); transient vs. sustained
<code>spend_z</code> , <code>mix_share_z</code> , <code>brand_spend_z</code> , <code>spend_vs_brand</code>	budget <i>reallocated</i> (brand total flat) vs. <i>removed</i> (total down)
<code>ctr_z</code> , <code>picr_z</code> , <code>cpc_z</code> (sign-flipped)	genuine efficiency gain / creative refresh
<code>impr_share_z</code>	survivorship bias (drops with the metric spike)
<code>brand_coincidence</code> , <code>market_coincidence</code>	external demand vs. brand-specific
<code>shape_sudden</code>	creative refresh/survivorship (sudden) vs. gain/mix shift (gradual)

Lag structure: features are contemporaneous with the flagged month - this diagnoses an already-detected spike, not a future one. The structural lag matters more: **every** quantity scoring month t uses only data strictly before t , the pooled seasonal index and peer sigma included. That is enforced, not asserted - `src/test_causality.py` re-scores a truncated panel and requires bit-identical results. An earlier version failed it: pooling those two over the full panel shifted 66% of month- t scores (max 7.3 z, against a ~ 1 z threshold); fixing it lowered the reported numbers and made them deployable.

5. Evaluation

Detector. Four-fold brand-grouped evaluation tunes the threshold on six brands and scores it on two unseen brands, rotating until every brand is held out once. Aggregated held-out performance is **precision 0.46** / **recall 0.76**. Only after that estimate is produced is the deployment threshold fit on all eight brands; `output/detector_brand_cv_metrics.csv` contains every fold.

Attribution, without clean labels in the real world. Nested cross-validation is grouped by *event*, not row. One event spans up to 3 months x several channels, so row-wise splitting leaks near-duplicates. Median imputation is fitted inside each fold, and calibration uses inner group-held-out predictions. **Effective sample size is events, not rows.**

Out-of-fold results (`output/attribution_metrics.csv`); lift is included because PR-AUC is not comparable across causes of differing prevalence.

cause	events	PR-AUC	base	lift
Survivorship bias	11	0.90	0.030	29.6x
Spend reduction artifact	8	0.21	0.024	8.7x
Creative refresh	21	0.65	0.079	8.2x
Genuine efficiency gain	15	0.58	0.084	6.9x
Mix shift artifact	6	0.30	0.056	5.3x
External demand spike	6	0.88	0.258	3.4x

Top-1 hit rate is **0.68**. PR-AUC is reported beside base rate because raw PR-AUC is not comparable across causes. Nested Platt scaling improves held-out Brier score for five causes; external demand worsens from 0.057 to 0.064 on only six events, a visible calibration-risk warning rather than a claim of certainty.

The training population is the serving population - flagged months, with **both** classes restricted. That symmetry is load-bearing. An intermediate version restricted only the negatives, keeping every labeled positive because a tiny label set could ill afford losing 37% of it. It looked excellent (mix-shift 2.21x, nothing at chance) and was an artifact: 37.5% of positives were unflagged vs 0% of negatives, and since `roas_sudden_z` is a feature and `flagged` is a threshold on it, “below the threshold” is “positive”. Unflagged-ness alone scored 2.33x - the entire headline. When a model runs downstream of a filter, both classes must come from the filtered population, and a jump on your weakest class is evidence of a leak before it is evidence of a fix.

`brand_spend_z` / `spend_vs_brand` separate budget *removed* (brand total falls) from *reallocated* (flat). The shared category-demand generator now makes cross-brand coincidence a real signal rather than a narrative unsupported by the data-generating process.

The synthetic-label model is only a cold start; the production evaluation loop is analyst-agreement rate over time (see *Productionisation*).

Predicted likelihoods are independent, not normalized contributions. Narratives include only causes clearing the 25% confidence bar; a weak second-ranked cause is omitted rather than promoted.

6. Causal honesty

Causal by construction: the *true* incrementality factor behind iROAS - never fully observed even in the synthetic data; the system sees a measured, demand-confounded proxy. **Associational, and presented as such:** ROAS, RROI, and every attribution output. The model is a differential-diagnosis tool, not a causal-inference engine - “consistent with a creative refresh,” never “the refresh caused this,” a distinction that lives in the narrative templates (`src/attribution.py`), not just here. **For a non-technical audience:** every output pairs a calibrated marginal likelihood with its evidence - “creative refresh (28%): CTR/PICR jumped” - and says “flagged, unexplained” when nothing is confident. A real causal claim needs a designed experiment (geo holdout, PSA test) or a validated MMM - neither runs here.

7. Productionisation

Retraining: monthly, on a rolling 24-36 month window. **Monitoring:** analyst agreement rate on attribution (the real accuracy proxy); feature drift on leading indicators (PSI/KS, monthly); alert-volume drift; raw-vs-calibrated Brier and lift-over-base per cause, so poor calibration or a cause degrading toward chance surfaces instead of looking authoritative. **Cold start:** exercised here (one brand launches mid-panel) - new brand-channels borrow the peer trend and sigma until a 3-month own-history floor. Next step, not built: cluster brands into spend/mix archetypes so new brands borrow from the nearest archetype, not the full pool. **Human-in-the-loop:** every analyst confirm/correct/reject is logged and feeds the next retrain - how the synthetic-label cold start gets replaced by real signal.

8. Limits - where this breaks

The simulator’s assumptions are the ceiling: if real response curves, seasonality, or confounding differ materially, both models need re-validation first. Event counts remain tiny (6-21 per cause), and external-demand calibration is unstable despite good discrimination. Short-history brands underperform on peer priors; the causal seasonal index is blind for a panel’s first 12 months. Attribution has no “none of the above” - a driver outside the taxonomy (competitor stockout, macro shock) yields uniformly low probabilities, surfaced as “flagged, unexplained.” Forecasting isn’t built. Best next inputs: analyst-confirmed labels and placement-level spend/revenue.